PROJECT REQUIREMENT DOCUMENT

Kreeda-Prerana Scout

Android App Development using Gen AI · Project #88

Domain: **Sports** | Category: **Grassroots Talent Discovery**

Submitted by	Email	Date
Sundar S	sundar.sh18@gmail.com	27 April 2026
University	**Team**	**Group**
VTU	Risk-Takers	G1

1. Executive Summary

Kreeda-Prerana Scout is a grassroots sports-talent tracking Android application designed for physical education teachers in rural schools. The app functions as a **Digital Scout**—enabling teachers to create athlete profiles, log performance in standard athletic tests with a built-in stopwatch, award automated milestone badges, and maintain an internal school leaderboard. Over successive seasons, the accumulated data creates a **Talent Curve** that makes it easy to identify students with state- or national-level potential, ensuring no rural talent goes unnoticed.

2. Problem Statement

Talented young athletes in rural schools often remain invisible to scouts and state-level selectors. The root cause is the absence of any structured, historical record of physical milestones such as sprint times, jump heights, and throw distances. Without a data trail:

- Coaches rely on subjective memory rather than objective numbers.
- Inter-school comparisons cannot be made systematically.
- Students who improve gradually are overlooked because no trend data exists.
- Pro-Kabaddi or state athletics scouts have no verifiable record to evaluate.

3. Vision & Objectives

The vision of Kreeda-Prerana is to democratise sports scouting by digitising the physical education process at its most fundamental level — the school trial session.

Key Objectives

#	Objective	Outcome
O1	Create structured athlete profiles	Unique identity per student-athlete
O2	Log trial results with precision timer	Objective, time-stamped performance data
O3	Auto-award milestone badges	Motivational recognition tied to benchmarks
O4	Generate school leaderboard	Healthy competition and ranking visibility
O5	Visualise talent curve over time	Trend graph for scout-ready comparison

4. Target Users

User Role	Description	Primary Actions
PE Teacher / Coach	Primary user; manages athlete data and trials	Create profiles, log trials, award badges, view leaderboard
School Administrator	Views aggregated school-wide talent summaries	View reports, export summaries
Student-Athlete	Passive beneficiary; motivated by badges & ranking	View own performance card & badges
Sports Scout (Future)	External user who may view exported performance cards	View / download shareable Performance Card image

5. Functional Requirements

5.1 Athlete Profile Management

- Create a new athlete profile with: Full Name, Age, Class/Standard, School, Primary Sport.
- Attach a profile photo using CameraX or Gallery picker.
- Edit or archive existing profiles without data loss.
- Search profiles by name or sport type.

5.2 Trial Logger with Precision Stopwatch

- Select the athlete and event type (100m Sprint, Long Jump, Shot Put, 400m, etc.).
- Built-in Chronometer accurate to two decimal places (00:00.00 format).
- Support manual distance/height input for field events (e.g., long jump in metres).
- Allow "undo last entry" to correct accidental taps.
- Batch Entry Mode: log results for an entire class of up to 30 students in a single session.
- Each entry is time-stamped and stored against the athlete's profile.

5.3 Milestone Badge System

- Pre-configured benchmark thresholds per event and age group (e.g., 100m < 13s = School Level Ready).
- App auto-awards badges upon crossing a threshold: School Level, District Level Ready, State Potential.

- Badge is persisted and displayed prominently on the athlete's profile card.
- Coach can view a badge audit trail with the date and trial that triggered it.

5.4 School Leaderboard

- Ranked list of athletes sorted by composite score across all events.
- Filter leaderboard by sport, age group, or class.
- Highlight top-3 performers with a distinct colour indicator.

5.5 Talent Curve (Performance Graph)

- Line graph showing a selected athlete's performance trend across multiple trial dates.
- Separate trend lines for each event type.
- Visual markers for each milestone badge earned on the timeline.
- Graph is clear, readable, and interpretable by a coach without data-science knowledge.

5.6 Shareable Performance Card

- Generate a single-screen image card containing athlete name, photo, top stats, and badges.
- Share card via WhatsApp, email, or any installed social app using Android Share Intent.
- Card must include school name and date of last trial.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Dashboard loads within 2 seconds on a mid-range device
NFR-02	Accuracy	Stopwatch accurate to ±0.01 seconds
NFR-03	Offline First	All data stored locally in Room DB; no internet required
NFR-04	Scalability	Supports up to 500 athlete profiles per installation
NFR-05	Usability	PE teachers with minimal tech literacy can log a trial in ≤3 taps
NFR-06	Data Safety	No data loss on app crash or phone restart
NFR-07	Architecture	MVVM pattern with clean separation of UI and business logic
NFR-08	Language	UI supports English (Kannada toggle as stretch goal)
NFR-09	Compatibility	Minimum SDK: Android 8.0 (API 26); Target SDK: Android 14

7. App User Flow

Step	Screen / Action	Description
Step 1	Launch	App opens to Dashboard showing total athletes, recent trials, and top badge earners.
Step 2	Athlete Onboarding	Coach taps '+' > enters name, age, sport, class > captures photo > saves profile.
Step 3	Start a Trial Session	Coach selects athlete > taps 'Log Trial' > chooses event type > starts Chronometer.
Step 4	Record Result	Coach stops timer or enters distance/height > result is auto-saved with timestamp.
Step 5	Batch Entry	For an entire class, coach uses Batch Mode: swipe through student list > enter each resul

Step 6	Badge Auto-Award	App compares result against benchmark thresholds > if exceeded, badge is displayed and
Step 7	View Talent Curve	Coach opens athlete profile > taps 'Trend' tab > views performance graph over time.
Step 8	Leaderboard Check	Coach taps 'Leaderboard' > filters by sport or class > views ranked list.
Step 9	Share Performance Card	Coach taps 'Share' on athlete profile > card is generated > share intent opens.

8. Technical Architecture & Implementation Hints

Component	Technology / Library	Purpose
UI Framework	Jetpack Compose / XML + ViewBinding	Declarative, responsive UI
Architecture	MVVM (ViewModel + LiveData / StateFlow)	Separation of concerns
Local Database	Room DB (Entities: Athlete, Trial, Badge)	Offline-first data persistence
Timer	Android Chronometer (System.nanoTime)	High-precision stopwatch (2 d.p.)
Charts	MPAndroidChart – LineChart	Talent Curve visualisation
Image Handling	CameraX + Coil / Glide	Profile & work photos
Share Intent	Android Bitmap + Intent.ACTION_SEND	Performance card sharing
Sorting / Ranking	Kotlin sortedByDescending / compareTo	Leaderboard generation
DI (optional)	Hilt / Koin	Dependency injection
Min SDK	API 26 (Android 8.0)	Broad device compatibility

Suggested Room DB Schema

Table	Key Columns
`athlete`	`id (PK), name, age, school, sport, photo_uri, created_at`
`trial`	`id (PK), athlete_id (FK), event_type, result_value, unit, trial_date`
`badge`	`id (PK), athlete_id (FK), badge_name, awarded_on, trial_id (FK)`

9. Sample Milestone Badge Benchmarks (Under-17 Category)

Event	School Level Ready	District Level Ready	State Potential
100 m Sprint	< 14.0 s	< 12.5 s	< 11.5 s
Long Jump	> 3.5 m	> 4.5 m	> 5.5 m
Shot Put	> 6.0 m	> 8.0 m	> 10.0 m
400 m Run	< 75 s	< 65 s	< 58 s
High Jump	> 1.1 m	> 1.3 m	> 1.5 m

Note: These benchmarks are indicative and configurable by the coach within the app settings.

10. Success Criteria

#	Criterion	Verification Method
SC-1	Timer accurate to two decimal places	Manual test: measure 10 s with phone stopwatch; delta ≤ 0.05 s
SC-2	Batch Entry supports 30 students per session	Create 30 dummy profiles; log results — no lag or crash
SC-3	Talent Curve graph is clear and interpretable	Peer review: 3 non-developer users can read trend without instructions
SC-4	Undo last trial entry works correctly	Log trial, tap undo, verify DB entry removed
SC-5	Performance Card shareable as image	Share to WhatsApp; card renders fully and correctly
SC-6	App loads main dashboard within 2 seconds	Profile with 100 athletes: measure cold-start time with Android Studio Profile
SC-7	Badge auto-awarded upon threshold breach	Unit test: input result exceeding benchmark; assert badge record in DB

11. Impact Goals

- **Khelo India Support:** Identifying rural talent early to feed into district and national academies.
- **Physical Literacy:** Encouraging a culture of fitness and systematic record-keeping in schools.
- **Equal Opportunity:** Ensuring a child in a remote village has a verified digital record of their talent, just like an urban athlete.
- **Data-Driven Coaching:** Moving PE teachers from memory-based coaching to evidence-based decisions.
- **Discipline & Motivation:** Badges and leaderboards incentivise students to train harder.

12. Assumptions & Constraints

Assumptions

- The app runs on Android smartphones owned by the PE teacher or school.
- No real-time internet connection is required for core functionality.
- Benchmark thresholds are pre-configured and adjustable by the teacher.
- Student photos are optional but encouraged for profile recognition.

Constraints

- Scope is limited to physical education trial events; team sports tactics are out of scope.
- The first release will not include a cloud-sync or multi-school comparison feature.
- Video capture and analysis are out of scope for Version 1.0.
- Actual Pro-Kabaddi scout integration is out of scope; the Performance Card serves as the bridge.

13. Suggested Development Timeline

Phase	Duration	Deliverable
Phase 1 – Setup & Data Layer	Week 1	Project setup, MVVM structure, Room DB schema, seed data
Phase 2 – Athlete Profile Module	Week 2	Create / edit / search athlete profiles with photo support
Phase 3 – Trial Logger & Stopwatch	Week 3	Chronometer, event selector, batch entry, undo functionality
Phase 4 – Badges & Leaderboard	Week 4	Badge auto-award logic, leaderboard screen with filters

Phase 5 – Talent Curve & Share	Week 5	MPAndroidChart integration, Performance Card bitmap + share intent
Phase 6 – Testing & Polish	Week 6	UI polish, edge-case testing, documentation, final APK build